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Inventor(s) CHEN; Liping et al.

## SIMPLIFIED SYSTEM AND METHOD FOR BIT-INTERLEAVED CODED MODULATION WITH ITERATIVE DECODING (BICM-ID)

### Abstract

Systems and methods for processing bit-interleaved coded modulation (BICM) signals from a BICM transmitter to generate information bit estimates of information in the BICM signals, including a decoder to generate the information bit estimates of the information in the received BICM signals and a symbol a posteriori probability (APP) generator to generate first symbol a posteriori probabilities (APPs) by processing the BICM signals based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator. The SPLLR generator generates the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), and the decoder generates the information bit estimates based on the first symbol APPs output from the symbol APP generator.

**Inventors:** **CHEN; Liping (Bethesda, MD), SESHADRI; Rohit Iyer (Gaithersburg, MD), BECKER; Neal David (Olney, MD), EROZ; Mustafa (Rockville, MD)**

**Applicant:** **HUGHES NETWORK SYSTEMS, LLC** (Germantown, MD)

**Family ID:** **92459022**

**Assignee:** **HUGHES NETWORK SYSTEMS, LLC (Germantown, MD)**

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### Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 18/366,318, filed Aug. 7, 2023, the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present application relates to simplified signal processing techniques for bit-interleaved coded modulation systems with iterative decoding (BICM-ID).

BACKGROUND

[0003] Techniques are known for iteratively exchanging soft-information between a binary LDPC (Low-Density Parity Check Code) decoder and symbol a posteriori probability generator in a receiver of a BICM system. Specifically, such iterative systems are known as bit-interleaved coded modulation with iterative decoding i.e., BICM-ID. These BICM-ID systems have the potential to improve the power efficiency of satellite links and this is an important enabler for techniques such as receiver-based interference cancellation. However, the conventional implementation complexity for current BICM-ID systems is quite high, and this has been a key impediment towards its adoption into state-of-the art receivers.

[0004] According to current ASIC (Application-specific integrated circuit) area estimation of BICM-ID implementation for current receivers, it is assumed that the Euclidean distance (ED) based symbol probability is generated at every BICM-ID outer iteration. This is done instead of calculating once and storing in memory since memory is more costly than logic in ASIC devices. Furthermore, the logic increase for BICM-ID operations is significant. In one implementation with only 24 parallel engines to generate the improved LLR (i.e., log likelihood ratio) input for a 360-engine LDPC decoder, the logic area will occupy approximately 95% of the current 360-engine DVBS2/X compatible LDPC decoder. This huge complexity increase from conventional satellite receivers prevents BICM-ID being a practical option in satellite receivers, even though it improves the spectral efficiency for the non-gray mapped constellations.

[0005] By way of background, it is imperative for state-of-the art wireless communication systems to operate with a high degree of efficiency to meet relentless data throughput demands. Furthermore, it is necessary to do this while contending with the conflicting realities of a very congested radio spectrum, the desire for low power-usage and the pragmatic need to keep hardware cost and complexity manageable. To quantify the trade-off between bandwidth and power efficiency, communication engineers have traditionally relied on a metric known as Shannon's channel capacity which helps determine the maximum spectral efficiency or rate at which information can be transmitted reliably through a channel, for some power (specifically, a signal power-to-noise power ratio (SNR)) requirement. The Shannon capacity can be shown as a function of SNR for the case of a transmitter communicating with a receiver over an additive white Gaussian noise (AWGN) channel and both employing only a single antenna (SISO). As an example, transmitting information at the rate of 2 bits-per-symbol reliably through an AWGN channel requires at least 5 dB SNR. Conversely, if the available SNR is 15 dB, it is not possible to extract more than 5 bits-per-symbol through the SISO-AWGN channel.

[0006] Shannon's limit, although groundbreaking, is an optimistic indicator of system performance since it assumes the use of Gaussian distributed modulation symbols. Practical systems, however, utilize symbols that are drawn with uniform probability from two-dimensional constellations having M-complex symbols such as QPSK (M=4), 8PSK (M=8), 16QAM (M=16) etc. In such scenarios, a better metric is the modulation constrained capacity, which measures the mutual information between the channel input and the channel output, under the constraint of utilizing uniformly distributed symbols belonging to two-dimensional constellations, such as those found in widely adopted standards such as DVB-S2 and DVB-S2X. This boundary can be determined for different practical modulation schemes and indicates a loss relative to the Shannon limit, especially at higher SNRs and increasing spectral efficiencies.

[0007] Pragmatic, state-of-the-art communication systems, such as 5G NR, DVB-S2/X employ powerful binary LDPC codes for forward error correction. In such systems, the information bits are encoded by the LDPC code and interleaved by a bit-interleaver. The interleaved bit-sequence is next mapped to one of M-possible symbols using a bit-to-symbol mapping rule such that the achievable performance also depends not only on the symbol constellation, but also on the mapping rule. This paradigm is known as Bit-interleaved coded modulation (BICM). The relevant performance limit for BICM systems is known as the pragmatic modulation capacity. For QPSK, the pragmatic modulation capacity is identical to the modulation constrained capacity. In contrast, there is a 0.25 dB SNR loss relative to the modulation constrained capacity for 4+12 APSK. This is because it is possible to realize perfect Gray bit-to-symbol mapping for QPSK thus ensuring a Hamming distance of 1 between the closest constellation symbols, i.e. their bit-to-symbol mapping label differs by exactly one bit. For the case of 4+12APSK and other power efficient APSK constellations, perfect Gray-labelling is generally not attainable, resulting in an SNR penalty with BICM.

SUMMARY

[0008] A receiver is provided for receiving and processing bit-interleaved coded modulation (BICM) signals from a BICM transmitter to generate information bit estimates of information in the BICM signals, the receiver including a decoder configured to generate the information bit estimates of the information in the received BICM signals, a symbol a posteriori probability (APP) generator configured to generate first symbol a posteriori probabilities (APPs) by processing the BICM signals based on Euclidean distances computed only for a quadrant to which received symbols derived from the BICM signals belong, and further based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator, wherein the extrinsic-information-based SPLLR generator is configured to generate the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), and the decoder is configured to generate the information bit estimates based on the first symbol APPs output from the symbol APP generator.

[0009] A method for receiving and processing bit-interleaved coded modulation (BICM) signals from a BICM transmitter to generate information bit estimates of information in the BICM signals, the method including generating, via a decoder, the information bit estimates of the information in the received BICM signals, generating, via a symbol a posteriori probability (APP) generator, first symbol a posteriori probabilities (APPs) by processing the BICM signals based on Euclidean distances computed only for a quadrant to which received symbols derived from the BICM signals belong, and further based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator, wherein the extrinsic-information-based SPLLR generator is configured to generate the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), and the decoder is configured to generate the information bit estimates based on the first symbol APPs output from the symbol APP generator.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements. Furthermore, it should be understood that the drawings are not necessarily to scale.

[0011] FIG. **1** illustrates a block diagram of a transmitter for transmitting BICM signals in a satellite system in accordance with an implementation of the disclosure.

[0012] FIG. **2** illustrates an example of a BICM-ID working model used as a prototype in developing an improved BICM-ID receiver in accordance with an implementation of the disclosure.

[0013] FIG. **3** illustrates an improved BICM-ID receiver developed from the prototype shown in FIG. **2** in accordance with an implementation of the disclosure.

[0014] FIGS. **4**A and **4**B show a constellation mapping and a constellation index, respectively, for operations of the receiver of FIG. **2** for an 8PSK system in accordance with an implementation of the disclosure.

[0015] FIGS. **5**A and **5**B illustrate a constellation mapping and a constellation index, respectively, for a system using the receiver of FIG. **3** for a 4+12APSK system in accordance with an implementation of the disclosure.

[0016] FIG. **6** illustrates a constellation that can be used to simplify the soft decision calculations for a 32APSK system in accordance with an implementation of the disclosure.

[0017] FIG. **7** illustrates a constellation that can be used to simplify the soft decision calculations for a 64APSK system in accordance with an implementation of the disclosure.

[0018] FIG. **8** illustrates a table of the differing complexities of the soft decision calculations using traditional BICM-ID receivers and the simplified BICM-ID receiver implementation of FIG. **3**, in accordance with an implementation of the disclosure.

[0019] FIG. **9** illustrates a flowchart of operations of the receiver shown in FIG. **3**.

[0020] FIG. **10** is a block diagram of an example computing device, which may be used to provide implementations of the mechanisms described herein.

DETAILED DESCRIPTION

[0021] In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. It will be apparent to persons of ordinary skill, upon reading this description, that various aspects can be practiced without such details. In other instances, well known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.

[0022] One method to mitigate the performance loss due to imperfect Gray mapping in BICM is by iteratively exchanging bit-level extrinsic

information between the LDPC decoder and the symbol a posteriori probability (APP) generator at the receiver. Such a process is known as bit-interleaved coded modulation with iterative decoding (BICM-ID). The block diagram of a working model BICM-ID system used as a prototype for developing the improved system described below (with respect to FIG. **3**) is shown in FIGS. **1** and **2**. Specifically, a transmitter **100** to provide BICM signals is shown in FIG. **1** and a working model (e.g., a prototype) of a receiver **200** for decoding the transmitted BICM signals is shown in FIG. **2**. An improved receiver **300** in accordance with the present disclosure is shown in FIG. **3** and will be discussed in detail below after the following discussion regarding the working model of FIG. **2**. The receiver **300** can also operate with BICM signals provided by the transmitter **100** shown in FIG. **1**.
[0023] As shown in FIG. **1**, a transmitter **100** transmits a data signal comprised of information bits over a communication channel **50** to a receiver **200** (or receiver **300** of FIG. **3**). The data signal may carry image, video, audio, and/or other information. Although a single channel **50** of communication is shown in this example, it should be appreciated that the technology described herein may be implemented in multi-channel communication systems. Additionally, although some example components of a transmitter **100** and receivers **200** and **300** are illustrated in this example, it should be appreciated that other transmitter and/or receiver configurations can be implemented, the order of components can be varied, some components may be excluded, added, or combined, and that one or more of these components can be implemented in either digital form (e.g., as software running on a DSP or other processing device, with the addition of a DAC) or as analog components.
[0024] Transmitter **100** includes an LDPC encoder **110**, interleaver **120**, a bit-to-symbol mapper **130** and a modulator **140**. In satellite communication system implementations, transmitter **100** may be a transmitter of a user terminal, such as, for example, a very small aperture terminal in (VSAT) that transmits on an inroute of the satellite communication system. Alternatively, transmitter **100** may be a transmitter of a satellite gateway that transmits on an outroute.
[0025] The information bits provided to the LDPC encoder **110** may include, for example, images, video, audio, text and other data. Although the information bits can be from a bit source which is separate from transmitter **100**, in some implementations, the bit source for the information bits may be incorporated into transmitter **100**. The information bits provided to the LDPC encoder **110** can be encapsulated to form baseband data frames or data blocks.
[0026] LDPC encoder **110** performs forward error correction (FEC) by adding redundancy to information data bits of the input signal. FEC can improve the reliability of transmission by adding redundant information to the data being transmitted through the channel. This redundant information is used to correct for errors introduced by the transmission of the signal over the transmission channel or link during signal reception. Examples of forward error correction codes that can be applied by LDPC encoder **110** can include block codes (e.g., low-density parity check codes (LDPC), turbo codes, Reed-Solomon codes, Hamming codes, Hadamard codes, BCH codes, and so on), and convolutional codes. In one particular implementation, the LDPC encoder **110** can be located at the output of a Bose-Chaudhuri-Hocquenghem codes (BCH) encoder (not shown) followed by the LDPC encoder **110**. The BCH encoder can process baseband data frames by adding additional redundant information based on one or more of the BCH encoding algorithms used in conjunction with the signal transmission protocol, and the LDPC encoder **110** can further processes the BCH encoded data frames to add a second layer of redundant information, or error correction information, to the data frames for error correction using one or more LDPC algorithms.
[0027] Interleaver **120** rearranges the encoded data bits to make distortion at receiver **200** (or receiver **300** of FIG. **3**) more independent from bit to bit and provide additional error correction capability during reception. For example, sections (e.g., bits) of data frames including one or more layers of error correction can be rearranged with respect to location or position (i.e., in time) within the data frames.
[0028] The bit-to-symbol mapper **130** and the modulator **140** operate together to map the interleaved bits from the interleaver **120** (e.g., within the frames) to complex-valued modulation symbols representing positions (i.e., in amplitude, phase, and/or time) in a modulated signal waveform based on a symbol constellation map for a modulation scheme. The bits can be modulated using any of a number of different modulation techniques by the modulator **140**. Examples of modulation schemes that can be implemented include amplitude and phase shift keying (APSK), e.g., 16-APSK, 32-APSK, 64-APSK, 128-APSK, or 256-APSK, quadrature phase shift keying (QPSK), 8PSK, M-ary phase shift keying (MPSK), quadrature amplitude modulation (QAM), and so forth.
[0029] Referring next to FIG. **2**, the receiver **200** includes a demodulator **210** for demodulating the received BICM signal from the transmitter **100** The demodulated received BICM signals from the demodulator **210** are provided to a symbol a posteriori probability (APP) generator **230**, which, in turn provides an output to a bit-LLR (Log Likelihood Ratio) generator **240** (which provides soft decisions SD for decoding, as will be discussed below). The symbol a posteriori probability (APP) generator **230** is structured to determine Euclidean distances as part of a demapping process for the received BICM signal that has been mapped and modulated in the transmitter **100**. Alternately, Euclidean distances can be calculated by a separate Euclidean distance calculator (not shown), as discussed in Applicant's U.S. application Ser. No. 17/977,760, which is hereby incorporated by reference in its entirety. Other elements of the receiver **200**, which will be discussed in greater detail below, are a first subtractor **245**, a deinterleaver **250**, an LDPC decoder **260**, a second subtractor **265**, an interleaver **270** and a symbol probability (SP) generator **280**. It is noted that, although the present description uses and LDPC encoder and decoder as examples, the present description could be used for any encoder and decoder that can be used for BICM-ID communications.
[0030] Still referring to FIG. **2**, an input to the LDPC decoder **260** is bit a priori probability and the decoder **260** output is bit a posterior probability. These probabilities are usually expressed as ratios in the log domain, namely Log Likelihood Ratios (LLRs). To begin the iterations at the receiver **200**, the symbol a posteriori probability generator **230** and bit-LLR generator **240** generates bit-level log-likelihood ratios (LLR) as initial soft decisions based upon receiving the noisy signal. After deinterleaving of these soft decisions by deinterleaver **250** (and performing subtraction by the first subtractor **245** of any previous extrinsic information EI from the output of the LDPC decoder **260**) they are fed as a priori information to the decoder **260**.
[0031] As also shown in FIG. **2**, the LDPC decoder **260** uses this a priori information to not only form estimates of the information bits but also to generate a posteriori probability LLRs of the associated code bits. Extrinsic information EI for the symbol a posteriori probability generator **230** is obtained by subtracting the a priori information at the input of the decoder **260** from the a posteriori LLRs at the output of the decoder **260** using the second subtractor **265**. Specifically, this extrinsic information EI is passed through the bit interleaver **270** and provided to the symbol probability generator **280** to generate symbol probabilities SP to provide to the symbol APP generator **230**. At this point, one outer iteration has been finished. For the next outer iteration, the LLRs will be updated based on the feedback extrinsic information EI and fed to LDPC decoder **260** to generate further a posteriori information, and the process will continue until the termination criteria is met.
[0032] The following discussion provides detailed mathematical implementation features regarding the operations of the receivers shown in FIGS. **2** and **3**. To this end, the following notations used in BICM-ID algorithms are noted:

Max-log approximation: Logsumexp(xi)=ln Σ.sub.ie.sup.x.sup.i=x.sub.k+ln Σ.sub.ie.sup.x.sup.i.sup.x.sup.k≈max(x.sub.k)
[0033] The decoder output bit LLR is defined by
[00001] $LLR_i = \ln\frac{p(c_i = 0)}{p(c_i = 1)}, \; = \; > \mathrm{LLR}_i = \ln(p(c_i = 0)) - \ln(p(c_i = 1)) = LL_i^0 - LL_i^1$
$\ln(p(c_i = 0)) = LL_i^0 = -\ln(1 + e^{-LLR_i})$ and $\ln(p(c_i = 1)) = LL_i^1 = -\ln(1 + e^{+LLR_i})$ $LL_i^0 = -\ln(1 + e^{-LLR_i}) = -\ln(e^0 + e^{-LLR_i}) \approx -\max(0, -\mathrm{LLR}_i)$
$LL_i^1 = -\ln(1 + e^{LLR_i}) = -\ln(e^0 + e^{LLR_i}) \approx -\max(0, \mathrm{LLR}_i)$ $LL_i^0 = LLR_i + LL_i^1 \approx LLR_i - \max(0, \mathrm{LLR}_i)$
[0034] It should be noted that the max log approximation used here is just an example for implementation, and the same principles shown in this disclosure would apply if some other approximation was used, or even if exact log sum exp was used.
[0035] Given the number of 2.sup.K constellation points, {s(m), m=0, . . . , 2.sup.K−1}, to map K channel bits (c**0**, c**1**, . . . , cK−**1**) and the AWGN

SNR estimate; each of demodulated symbols {r(i), i=0, 1, . . . , N−1} are converted by the Euclidean distance calculator to 2.sup.K Euclidean distances scaled by SNR, given by {ED(i, m)=−SNR.Math.|r(i)−s(m)|.sup.2, i=0, 1, . . . , N−1, m=0, 1, . . . , 2.sup.K−1} before generating K soft decisioned bits, {SD(i, k), i=0, 1, . . . , N−1, k=0, 1, . . . , K−1}, per each of demodulated symbols.
[0036] After a given number of iterations, the second subtractor **265** generates the LLR extrinsic information, denoted as {EI(DEINT (i, k)), i=0, 1, . . . , N−1, k=0, 1, . . . , K−1}, which will be interleaved by interleaver **270** to match the order of demodulated symbols, {EI(i, k), i=0, 1, . . . , N−1, k=0, 1, . . . , K−1}. The Log Likelihood (LL) of the extrinsic information, denoted by BI(i, k, q), to be q=1 or 0 is converted from LLR, which is defined by:
[00002] $BI(i,k,1) = -LES(0, EI(i,k)), \mathrm{BI}(i,k,0) = EI(i,k) - LES(0, EI(i,k)),$ [0037] Where LES is logsumexp func or an approximation thereof.
[0038] The function of LES(x.sub.0, x.sub.1, . . . x.sub.k) is defined as follows:
[00003] $\mathrm{LES}(x_0, x_1, \text{.Math.}, x_k) = LES(LES(x_0, x_1, \text{.Math.}, x_{k-1}), x_k), \mathrm{LES}(x_0, x_1) = \max(x_0, x_1).$ [0039] (Assuming maxlog approximation for logsumexp)
[0040] According to DVBS2-X standard, soft decisioned (SD) bits are deinterleaved by deinterleaver **250** depending on the number of constellation points via row write and column read to be fed to the LDPC decoder **260** as {SD(DEINT(i, k)), where symbol index i=0, 1, . . . , N−1, bit index of the symbol k=0, 1, . . . , K−1}
[0041] The soft decision bits in the form of Maximum Likelihood (ML) LLR indicates the confidence of the demapping decision, which are computed by the symbol APP generator **230** with ED based symbol probabilities ED(i, m) as well as the extrinsic information from previous iteration based symbol probabilities SP(i, m), where received symbol index i=0, 1, . . . , N−1, constellation index m=0, 1, . . . , 2.sup.k−1}, SP(i, m) are initially all 0 at first BICM-ID iteration.
[00004] $\mathrm{LR}(i,k) = \ln \frac{\text{.Math.}_{s(m) \in A_{ck=0}} \exp(-\frac{\text{.Math. } r(i)-s(m) \text{ .Math.}^2}{SNR} + \text{.Math.}_{k=0}^{K-1} \mathrm{BI}(i,k,\mathrm{ck} \in s(m)))}{\text{.Math.}_{s(m) \in A_{ck=1}} \exp(-\frac{\text{.Math. } r(i)-s(m) \text{ .Math.}^2}{SNR} + \text{.Math.}_{k=0}^{K-1} \mathrm{BI}(i,k,\mathrm{ck} \in s(m)))} = \ln \frac{\text{.Math.}_{s(m) \in A_{ck=0}} \exp(ED(i,m) + SP(i,m))}{\text{.Math.}_{s(m) \in A_{ck=1}} \exp(ED(i,m) + SP(i,m))}$ [0042] where ck denotes the kth mapping bit of constellation point s(m), S.sub.m∈A.sub.ck=1 means the symbols belong to a subset of constellations, of which the kth mapping bit equals to 1. BI(i, k, ck∈s(m)) denotes the log likelihood of kth mapping bit of ith received symbol equals to the kth mapping bit of constellation s(m).
[0043] The usual implementation of the ML LLR or logsumexp will be reduced to max-log LLR, as follows:
[00005] $SD(i,k) = LLR(i,\ k) = \max_{s(m) \in A_{ck=0}}(ED(i,m) + SP(i,m)) - \max_{s(m) \in A_{ck=1}}(ED(i,m) + SP(i,m))$
[0044] For an example, in the case of 8PSK, 2.sup.K=8 constellation points, the i-th symbol of which carries K=3 channel bits representing the binary form of the constellation. The bit mapping of the constellation is shown in FIG. **4**A and constellation index is shown in FIG. **4**B.
[0045] For the ith symbol, ck denotes the kth soft decisioned (c**0**, c**1**, c**2** as shown in FIG. **4**A) bit LLRs defined by SD(i,k) are given by finding the maximum improved symbol a posteriori probability (ED+SP) of all the symbols with the kth bit to be 0 (noted by diamond-shaped constellations in FIG. **4**A) and to be 1 (noted by circular constellations in FIG. **4**B), then perform subtraction to obtain improved bit LLR:
[00006]
$\mathrm{SD}(i,0) = LES(ED(i,0) + SP(i,0), ED(i,3) + SP(i,3), ED(i,4) + SP(i,4), ED(i,7) + \mathrm{SP}(i,7))) - \mathrm{LES}(ED(i,1) + SP(i,1), \mathrm{ED}(i,2) + SP(i,2),\ ED(i,5) + S$
$\mathrm{SD}(i,1) = \mathrm{LES}(ED(i,0) + SP(i,0), ED(i,1) + SP(i,1), ED(i,6) + SP(i,6), ED(i,7) + \mathrm{SP}(i,7)) - \mathrm{LES}(ED(i,2) + SP(i,2),\ ED(i,3) + SP(i,3), ED(i,4) + SP$
$\mathrm{SD}(i,2) = \mathrm{LES}(ED(i,0) + SP(i,0), ED(i,1) + SP(i,1), ED(i,2) + SP(i,2), ED(i,3) + \mathrm{SP}(i,\ 3)) - \mathrm{LE}S(ED(i,4) + SP(i,4),\ ED(i,4) + SP(i,4), ED(i,6) + S$
[0046] For constellation index m=0, 1, . . . , K−1, EI based symbol probability SP(i, m), which defines the probability of the ith symbol to be mth constellation defined in FIG. **4**B, is calculated by sum up Log Likelihood probability of all the K bits pattern defined for the mth constellation, e.g. for symbol index to be 0, the bit pattern is 000, the probability of the received symbol r(i) to be constellation index 0 defined by SP(i, 0) is calculated by sum up the LL probability of all three mapping bits to be 0, BI(i, k, 0) for k=0, 1, 2.
[00007] $\mathrm{SP}(i,0) = \mathrm{BI}(i,0,0) + \mathrm{BI}(i,1,0) + \mathrm{BI}(i,2,0), c0c1c2 = \text{“}000\text{”}\ \mathrm{SP}(i,1) = \mathrm{BI}(i,0,1) + \mathrm{BI}(i,1,0) + \mathrm{BI}(i,2,0), c0c1c2 = \text{“}100\text{”}$
$\mathrm{SP}(i,2) = \mathrm{BI}(i,0,1) + \mathrm{BI}(i,1,1) + \mathrm{BI}(i,2,0), c0c1c2 = \text{“}110\text{”}\ P(i,3) = \mathrm{BI}(i,0,0) + \mathrm{BI}(i,1,1) + \mathrm{BI}(i,2,0), c0c1c2 = \text{“}010\text{”}$
$\mathrm{SP}(i,4) = \mathrm{BI}(i,0,0) + \mathrm{BI}(i,1,1) + \mathrm{BI}(i,2,1), c0c1c2 = \text{“}011\text{”}\ \mathrm{SP}(i,5) = \mathrm{BI}(i,0,1) + \mathrm{BI}(i,1,1) + \mathrm{BI}(i,2,1), c0c1c2 = \text{“}111\text{”}$
$\mathrm{SP}(i,6) = \mathrm{BI}(i,0,1) + \mathrm{BI}(i,1,0) + \mathrm{BI}(i,2,1), c0c1c2 = \text{“}101\text{”}\ \mathrm{SP}(i,7) = \mathrm{BI}(i,0,0) + \mathrm{BI}(i,1,0) + \mathrm{BI}(i,2,1), c0c1c2 = \text{“}001\text{”}$
[0047] The above discussion pertains to the operation of the receiver of FIG. **2**. Turning now to FIG. **3**, a receiver **300** is shown which was developed to significantly reduce the amount of calculation necessary in the receiver shown in FIG. **2** without adversely affecting the accuracy of the final information bit estimates. To this end, the receiver **300** of FIG. **3** includes a demodulator **310** for demodulating the received BICM signal from the transmitter **100** The demodulated signals are provided to an improved symbol a posteriori probability (APP) generator **330**, which, in turn provides an output to an improved bit-LLR (Log Likelihood Ratio) generator **340** (which provides soft decisions (SDs) for decoding, as will be discussed below). In a first outer iteration for each received symbol, the quadrant is determined by the received symbol r(i), and for the rest of the iteration, the quadrant is determined by the maximum extrinsic based symbol probability, either SP(i, m) or SPLLR(i, m), for m=0, 1, . . . 2K−1. This provides a method to decide which quadrant the received symbol belongs to.
[0048] The symbol a posteriori probability (APP) generator **230** is structured to determine Euclidean distances only for a quadrant which received signals might belong to as part of a demapping process for the received BICM signal that has been mapped and modulated in the transmitter **100**. Limiting the computation of Euclidean distances by focusing only on the quadrant to which the received symbols might belong is significant for further reducing complexity of the computations necessary for processing the received BICM signals. The determination of Euclidean distances is discussed in Applicant's U.S. application Ser. No. 17/977,760, which is hereby incorporated by reference in its entirety.
[0049] Other elements of the receiver **300** shown in FIG. **3**, which will be discussed in greater detail below, are a first subtractor **345**, a deinterleaver **350**, an LDPC decoder **360**, a second subtractor **365**, an interleaver **370** and an improved symbol probability (SP) generator **380**. The nature of the improvements implemented in the improved symbol APP generator (ED+SPLLR) **330**, the improved bit-LLR generator (SD) **340** and the improved EI-based symbol probability (SPLLR) generator **380** are discussed below.
[0050] Specifically, the present disclosure proposes a novel receiver and method which will greatly reduce the implementation complexity of BICM-ID receivers. The simplification is performed at the calculation of log-likelihood ratio (LLR) by the symbol APP generator **330** and the bit-LLR soft decision generator **340** to provide the input to the LDPC decoder **360**. The LLR calculation for BICM-ID receiver **300** will combine the channel input symbol probability which is based on Euclidean Distance (ED) and the decoder **360** feedback symbol probability which is based on extrinsic bit LLR of the decoder **360** output.
[0051] The feedback extrinsic information-based symbol probability calculation performed in the EI-based symbol probability (SPLLR) generator **380** can be simplified and, as will be shown below, does not involve any approximation, and hence no performance degradation. This is an advantage of using the feedback extrinsic based symbol probability calculation as a first simplification of the calculations in accordance with the present disclosure. The symbol probability based on ED can be further simplified by utilizing the symmetric properties of the constellation by only taking the information relative to only one quadrant of all the constellation, which will cause very small performance loss. This is also a significant feature as a second simplification of the present disclosure in greatly decreasing the implementation complexity. By combining both simplifications, the amount of mathematical operations for the bit-LLR calculation (soft decisions) provided at the output of the bit-LLR generator **340** can be reduced by approximately half in the receiver **300** of FIG. **3** in comparison with the receiver **200** of FIG. **2**.
[0052] As noted above, the receiver **200** of FIG. **2** converts decoder **260** output bit LLR to bit LL (log likelihood first, and then sums up the LL of the corresponding mapping bits pattern to get the LL probability SP(i, m) of all 2.sup.K constellation points, {s(m), m=0, . . . , 2.sup.K−1}. The math

expression is shown in the previous discussion as:

[00008]$SD(i,k) = LLR(i,k) = \max_{s(m) \in A_{ck=0}} (ED(i,m) + SP(i,m)) - \max_{s(m) \in A_{ck=1}} (ED(i,m) + SP(i,m))$

[0053] It has been determined that no substantial change will result to the numerical result of SD(i, k) if we subtract a common factor SP(i, n) from all the SP(i, m) in the above formula, where integer n can be any number from 0 to 2.sup.K−1. For convenience of notation, we choose n to be the constellation index, of which the corresponding bit pattern c.sub.0c.sub.1c.sub.2 . . . c.sub.K−1 are all ones, we arrive at:

[00009]

$SD(i,k) = \max_{s(m) \in A_{ck=0}} (ED(i,m) + SP(i,m) - SP(i,n)) - \max_{s(m) \in A_{ck=1}} (ED(i,m) + SP(i,m) - \mathrm{SP}(i,n)) = \; = \max_{s(m) \in A_{ck=0}} (ED(i,m) + SPLLR(i,$

[0054] Regarding the above discussion, it is noted that with regard to the symbol LLR:

[00010]$r_i, \mathrm{SPLLR}_i = \ln\frac{p(r_i = s(m))}{p(r_i = s(n))}, \mathrm{SP}(i,n) = \ln p(r_i = s(n))$

[0055] Keeping in mind the above discussion, BI(i, k, 1)=−LES(0, EI(i, k)), BI(i, k, 0)=EI(i, k)−LES(0, EI(i, k)), can derive that BI(i, k, 0)−BI(i, k, 1)=EI(i, k)

[00011]$\mathrm{SPLLR}(i,m) = SP(i,m) - \mathrm{SP}(i,n) = \; .\mathrm{Math.}_{k=0}^{K-1} \; \mathrm{BI}(i,k,\mathrm{ck} \in s(m)) - \; .\mathrm{Math.}_{k=0}^{K-1} \; \mathrm{BI}(i,k,\mathrm{ck} \in s(n)) = \; .\mathrm{Math.}_{\mathrm{ck} \in s(m), ck=0} \; \mathrm{EI}(i,k)$

[0056] The improved lossless way to calculate symbol LL probability SP(i, m) is to use LLR instead of LL in the EI-based symbol probability generator **380**. Firstly we calculate symbol LLR SPLLR(i, m) directly from bit LLR which is the output EI(i, k) of interleaver **370**. Using 8PSK as an example, constellation index n=5 corresponds to three mapping bits to be “111”, the SPLLR calculation for each constellation s(m) is as follows:

[00012]

$\mathrm{SPLLR}(i,0) = \mathrm{SP}(i,0) - SP(i,5) = BI(i,0,0) + BI(i,1,0) + \mathrm{BI}(i,2,0) - (BI(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,1)) = (\mathrm{BI}(i,0,0) - BI(i,0,1)) + (\mathrm{BI}(i,1,0) - BI(i,1,1)) +$

$\mathrm{SPLLR}(i,1) = SP(i,1) - SP(i,5) = \mathrm{BI}(i,0,1) + BI(i,1,0) + \mathrm{BI}(i,2,0) - (BI(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,1)) = \mathrm{BI}(i,1,0) - BI(i,1,1) + (\mathrm{BI}(i,2,0) - BI(i,2,1)) =$

$\mathrm{SPLLR}(i,2) = \mathrm{SP}(i,2) - SP(i,5) = BI(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,0) - (BI(i,0,1) + \mathrm{BI}(i,1,1) + \mathrm{BI}(i,2,1)) = BI(i,2,0) - \mathrm{BI}(i,2,1) = \mathrm{EI}(i,2)$

$\mathrm{SPLLR}(i,3) = P(i,3) - SP(i,5) = BI(i,0,0) + BI(i,1,1) + \mathrm{BI}(i,2,0) - (BI(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,1)) = (\mathrm{BI}(i,0,0) - BI(i,0,1)) + (\mathrm{BI}(i,2,0) - BI(i,2,1)) =$

$\mathrm{SPLLR}(i,4) = \mathrm{SP}(i,4) - \mathrm{SP}(i,5) = BI(i,0,0) + BI(i,1,1) + \mathrm{BI}(i,2,1) - (BI(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,1)) = (\mathrm{BI}(i,0,0) - BI(i,0,1)) = \mathrm{EI}(i,0)$

$\mathrm{SPLLR}(i,5) = SP(i,5) - SP(i,5) = 0$

$\mathrm{SPLLR}(i,6) = \mathrm{SP}(i,6) - SP(i,5) = BI(i,0,1) + BI(i,1,0) + \mathrm{BI}(i,2,1) - (BI(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,1)) = (\mathrm{BI}(i,1,0) - BI(i,1,1)) = \mathrm{EI}(i,1)$

$\mathrm{SPLLR}(i,7) = \mathrm{SP}(i,6) - SP(i,5) = BI(i,0,0) + BI(i,1,0) + BI(i,2,1) - (\mathrm{BI}(i,0,1) + \mathrm{BI}(i,1,1) + BI(i,2,1)) = (\mathrm{BI}(i,0,0) - BI(i,0,1)) + (\mathrm{BI}(i,1,0) - BI(i,1,1)) =$

[0057] For the 8PSK constellation as shown in FIGS. **4**A and **4**B, the symbol LLR of the constellation index m SPLLR(i, m) with respect to symbol index n=5 is just the sum of the bit LLRs for those three mapping bits which have a 0 in that bit position of constellation s(m). The final results below show that addition operation will only be needed for bit equal to 0, and that is half of all the addition for the current practice. In addition, there is no need to convert LLR to LL (or from EI to BI).

[00013]$c0c1c2 = 000, \mathrm{SPLLR}(i,0) = SP(i,0) - SP(i,5) = \mathrm{EI}(i,0) + \mathrm{EI}(i,1) + \mathrm{EI}(i,2)$

$c0c1c2 = 100, SPLLR(i,1) = SP(i,1) - SP(i,5) = \mathrm{EI}(i,1) + \mathrm{EI}(i,2) c0c1c2 = 110, \mathrm{SPLLR}(i,2) = SP(i,2) - SP(i,5) = \mathrm{EI}(i,2)$

$c0c1c2 = 010, SPLLR(i,3) = SP(i,3) - SP(i,5) = \mathrm{EI}(i,0) + \mathrm{EI}(i,2) c0c1c2 = 011, SPLLR(i,4) = SP(i,4) - SP(i,5) = \mathrm{EI}(i,0)$

$c0c1c2 = 111, SPLLR(i,5) = SP(i,5) - SP(i,5) = 0 c0c1c2 = 101, SPLLR(i,6) = SP(i,6) - SP(i,5) = \mathrm{EI}(i,1)$

$c0c1c2 = 001, SPLLR(i,7) = SP(i,6) - SP(i,5) = \mathrm{EI}(i,0) + \mathrm{EI}(i,1)$

[0058] Secondly, we replace the SP with SPLLR in the calculation of the soft decisions SD by the bit-LLR generator **340** so that the final calculations will be as follows:

[00014]

$\mathrm{SD}(i,0) = LES(ED(i,0) + SPLLR(i,0), ED(i,3) + SPLLR(i,3), ED(i,4) + \mathrm{SPLLR}(i,4), ED(i,7) + SPLLR(i,7)) - \mathrm{LES}(\mathrm{ED}(i,1) + \mathrm{SPLLR}(i,1), ED(i,2) +$

$\mathrm{SD}(i,\ 2) = LES(ED(i,0) + SPLLR(i,0), ED(i,1) + SPLLR(i,1) ED(i,\ 2) + SPLLR(i,\ 2), ED(i,3) + SPLLR(i,3) - LES(ED(i,\ 4) + SPLLR(i,\ 4), ED(i$

$\mathrm{SD}(i,1) = LES(ED(i,0) + SPLLR(i,0), ED(i,1) + \mathrm{SPLLR}(i,1), ED(i,6) + \mathrm{SPLLR}(i,6), ED(i,7) + SPLLR(i,7)) - \mathrm{LES}(ED(i,2) + \mathrm{SPLLR}(i,2), ED(i\,3) +$

$\mathrm{SD}(i,\ 2) = \mathrm{LES}(\mathrm{ED}(i,0) + SPLLR(i,0), ED(i,1) + SPLLR(i,1) ED(i,\ 2) + \mathrm{SPLLR}(i,2), ED(i,3) + \mathrm{SPLLR}(i,3) - \mathrm{LES}(ED(i,4) + SPLLR(i,4), ED(i,4) +$

[0059] In general, the complexity of SPLLR calculations will be less than half of the SP calculation for any constellation. Simplification can be achieved in two ways: (1) It is unnecessary to calculate BI(i, k, 1) and BI(i, k, 0); and (2) SPLLR calculation only use the needed EI information for the corresponding bit=0 in the constellation bit mapping.

[0060] It is noted that there is no performance loss in the simplification by replacing symbol probability (SP) with symbol probability log-likelihood ratio (SPLLR), and the complexity is still significant since it is still necessary to calculate ED(i, m)+SPLLR(i, m) for all the constellation points as well as the two multi-input maximum function. When constellation becomes large, e.g., 64APSK or 128APSK, the math operation will be overwhelming and complexity for updating SD could be equivalent to or larger than that of the LDPC decoder.

[0061] Symmetry of the constellation is leveraged next to further reduce complexity of Euclidean Distance (ED) computation by focusing on the quadrant to which the received symbol might belong. A 4+12APSK constellation shown in FIGS. **5**A and **5**B is used as an example. Four bits c**0**,c**1**,c**2**,c**3** will be used to map to one of the 16 constellation points, for bit c**0**, the constellation points as diamonds corresponds to bit c**0**=1 and the constellation points shown as circles corresponds to c**0**=0. The same rule applies to bit c**1**, c**2** and c**3**.

[0062] From FIG. **5**A, we can find the corresponding constellation index set CI(k, q) and CI(k, q), where binary bit q=0 and 1, and bit index k=0, 1, . . . K−1. [0063] q=0, k=0, constellation index set CI(0, 0)={4 15 7 8 9 10 13 14} [0064] q=0, k=1, constellation index set CI(1, 0)={4 5 6 7 10 11 12 13} [0065] q=0, k=2, constellation index set CI(2, 0)={0 4 5 15 3 12 13 14} [0066] q=0, k=3, constellation index set CI(3, 0)={0 4 5 15 1 6 7 8} [0067] q=1, k=0, constellation index set CI(0, 1)={0 5 1 6 2 11 3 12} [0068] q=1, k=1, constellation index set CI(1, 1)={0 15 1 8 2 9 3 14} [0069] q=1, k=2, constellation index set CI(2, 1)={1 6 7 8 2 9 10 11} [0070] q=1, k=3, constellation index set CI(3, 1)={2 9 10 11 3 12 13 14}

[0071] For the i-th received symbol of which carries 4 channel bits, 4 soft decisioned bits are given by:

[00015]

$\mathrm{SD}(i,0) = \mathrm{LES}(ED(i,4) + \mathrm{SPLLR}(i,4), ED(i,15) + \mathrm{SPLLR}(i,15), ED(i,7) + \mathrm{SPLLR}(i,7), ED(i,8) + \mathrm{SPLLR}(i,8), \mathrm{ED}(i,9) + \mathrm{SPLLR}(i,9), ED(i,10) + \mathrm{SPLL}$

$\mathrm{SD}(i,1) = \mathrm{LES}(ED(i,4) + \mathrm{SPLLR}(i,4), ED(i,5) + \mathrm{SPLLR}(i,5), \mathrm{ED}(i,6) + \mathrm{SPLLR}(i,6), \mathrm{ED}(i,7) + \mathrm{SPLLR}(i,7), \mathrm{ED}(i,10) + \mathrm{SPLLR}(i,10), ED(i,11) + \mathrm{SPLLR}$

$\mathrm{SD}(i,2) = LES(ED(i,0) + \mathrm{SPLLR}(i,0), ED(i,4) + \mathrm{SPLLR}(i,4), ED(i,5) + \mathrm{SPLLR}(i,5), ED(i,15) + \mathrm{SPLLR}(i,15), ED(i,3) + \mathrm{SPLLR}(i,3), ED(i,12) + \mathrm{S}$

$\mathrm{SD}(i,3) = \mathrm{LES}(ED(i,0) + \mathrm{SPLLR}(i,0), ED(i,4) + \mathrm{SPLLR}(i,4), ED(i,5) + \mathrm{SPLLR}(i,5), ED(i,15) + \mathrm{SPLLR}(i,15), ED(i,1) + \mathrm{SPLLR}(i,1), ED(i,6) + \mathrm{SPLLR}$

[0072] It is noted that the complexity of SD(i, k) calculation for 4+12APSK constellation has greatly increased compared with 8PSK. Since {ED(i, m)=−SNR.Math.|r(i)−s(m)|.sup.2, i=0, 1, . . . , N−1, m=0, 1, . . . , 2.sup.K−1}, max.sub.m(ED(i, m))), m∈CI(m, k) and is equivalent to min.sub.m(SNR.Math.|r(i)−s(m)|.sup.2). The distance defined by |r(i)−s(m)| will be smaller in the quadrant r(i) belongs than the rest of the quadrants.

[0073] Also, we can separate ED (i, m) into real and imaginary expression:

[00016]
$$\begin{aligned} \mathrm{ED}(i,\ m) &= -\mathrm{SNR}\ .\mathrm{Math.}\ \ .\mathrm{Math.}\ r(i) - s(m)\ .\mathrm{Math.}^{2} \\ &= -\mathrm{SNR}\ .\mathrm{Math.}\ ((\mathrm{real}(r(i) - s(m))^2 + \mathrm{imag}(r(i) - s(m))^2) \\ &= \mathrm{EDI}(i,m) + EDQ(i,m) \end{aligned}$$

For mapping bits C**0** and C**1** as shown in FIG. **5**A, we will only calculate the ED(i, m) in the quadrant of interest which will reduce the amount of ED calculations as well as the number of entries of max function from 8 to 2. For bit C**2**, we can only calculate the distance using real part EDI(i, m) instead of complex number. For bit C**3**, we can only calculate the distance using imaginary part EDQ(i, m) instead of complex number. These ED calculations may be performed by the symbol APP generator **330**, as discussed above.

[0074] The simplified or reduced SD calculation in the bit-LLR generator **340** of 4 soft decisioned bits are given as follows according to different quadrants. In a first outer iteration for each received symbol, the quadrant is determined by the received symbol r(i), and for the rest of the iteration the quadrant is determined by the maximum extrinsic based symbol probability, either SP(i, m) or SPLLR(i, m), for m=0, 1, . . . , 2.sup.K−1. If the received symbol is located in first quadrant, only the constellation point with index m= {0, 4, 5, 15} will be used to update the SD for bit C**0** and C**1**; the constellation point with index m= {0, 4, 5, 15, 1, 6, 7, 8} will be used for bit C**2** and only real part will be used for ED calculation; the constellation point with index m= {0, 4, 5, 15, 3, 12, 13, 14} will be used for bit C**3** and only imaginary part will be used for ED calculation. The complexity of computation will be reduced significantly as shown below.

[0075] If first quadrant:

[00017]$SD(i,0) = LES(ED(i,4) + SPLLR(i,4), ED(i,15) + SPLLR(i,15)) - LES(ED(i,0) + SPLLR(i,0), ED(i,5) + SPLLR(i,5)))$

$SD(i,1) = LES(ED(i,4) + SPLLR(i,4), ED(i,5) + SPLLR(i,5)) - LES(ED(i,0) + SPLLR(i,0), ED(i,15) + SPLLR(i,15))$

$SD(i,2) = LES(EDI(i,0) + SPLLR(i,0), EDI(i,4) + SPLLR(i,4), EDI(i,5) + SPLLR(i,5), EDI(i,15) + SPLLR(i,15)) - LES(EDI(i,1) + SPLLR(i,1), EDI$

$SD(i,3) = LES(EDQ(i,0) + SPLLR(i,0), EDQ(i,4) + SPLLR(i,4), EDQ(i,5) + SPLLR(i,5), EDQ(i,15) + SPLLR(i,15) - LES(EDQ(i,3) + SPLLR(i,3), E$

[0076] If second quadrant:

[00018]$SD(i,0) = LES(ED(i,7) + SPLLR(i,7), ED(i,8) + SPLLR(i,8)) - LES(ED(i,1) + SPLLR(i,1), ED(i,6) + SPLLR(i,6))$

$SD(i,1) = LES(ED(i,6) + SPLLR(i,6),\ ED(i,7) + SPLLR(i,7)) - LES(ED(i,1) + SPLLR(i,1), ED(i,S) + SPLLR(i,8))$

$SD(i,2) = LES(EDI(i,0) + SPLLR(i,0), EDI(i,4) + SPLLR(i,4), EDI(i,5) + SPLLR(i,5), EDI(i,15) + SPLLR(i,15) - LES(EDI(i,1) + SPLLR(i,1), EDI$

$SD(i,3) = LES(EDQ(i,1) + SPLLR(i,1), EDQ(i,6) + SPLLR(i,6), EDQ(i,7) + SPLLR(i,7), EDQ(i,8) + SPLLR(i,8)) - LES(EDQ(i,2) + SPLLR(i,2$

[0077] If third quadrant:

[00019]$SD(i,0) = LES(ED(i,9) + SPLLR(i,9), ED(i,10) + SPLLR(i,10)) - LES(ED(i,2) + SPLLR(i,2), ED(i,11) + SPLLR(i,11))$

$SD(i,1) = LES(ED(i,10) + SPLLR(10), ED(i,11) + SPLLR(i,11)) - LES(ED(i,2) + SPLLR(i,2), ED(i,9) + SPLLR(i,9))$

$SD(i,2) = LES(EDI(i,3) + SPLLR(i,3), EDI(i,12) + SPLLR(i,12), EDI(i,13) + SPLLR(i,13), EDI(i,14) + + SPLLR(i,14)) - LES(EDI(i,2) + SPLLR(i,2), EDI$

$SD(i,3) = LES(EDQ(i,1) + SPLLR(i,1), EDQ(i,6) + SPLLR(i,6), EDQ(i,7) + PLLR(i,7), EDQ(i,8) + SPLLR(i,8) - LES(EDQ(i,2) + SPLLR(i,2), E$

[0078] If fourth quadrant:

[00020]$SD(i,0) = LES(ED(i,13) + SPLLR(i,13), ED(i,14) + SPLLR(i,14)) - LES(ED(i,3) + SPLLR(i,3), ED(i,12) + SPLLR(i,12))$

$SD(i,1) = LES(ED(i,12) + SPLLR(i,12), ED(i,13) + SPLLR(i,13) - LES(ED(i,3) + SPLLR(i,3), ED(i,14) + SPLLR(i,14))$

$SD(i,2) = LES(EDI(i,3) + SPLLR(i,3), EDI(i,12) + SPLLR(i,12), EDI(i,13) + SPLLR(i,13), EDI(i,14) + SPLLR(i,14)) - LES(EDI(i,2) + SPLLR(i,2), EDI$

$SD(i,3) = LES(EDQ(i,0) + SPLLR(i,0), EDQ(i,4) + SPLLR(i,4), EDQ(i,5) + SPLLR(i,5), EDQ(i,15) + SPLLR(i,15)) - LES(EDQ(i,3) + SPLLR(i,3), EDQ$

[0079] Other constellations, as shown in FIG. **6** and FIG. **7** for 32APSK and 64APSK, respectively, share the same symmetry and can also be adapted to simplify the SD calculations. Only the real part is used for bit C**2** and the imaginary part is used for bit C**3**, only one quadrant is used for the rest of the mapping bits. These simplified calculations can be performed in the symbol APP generator **330**. The table shown in FIG. **8** will summarize the savings by utilizing the constellation symmetry for different constellations. Comparing the top and bottom part of the table, we can see the complexity can be reduced by about 50%.

[0080] FIG. **8** shows a table of comparisons between the implementation complexities of the operations using the traditional BICM-ID receivers and the simplified BICM-ID receiver implementation shown in the receiver of FIG. **3**. Since there is some information loss in the simplifications achieved by the receiver of FIG. **3**, there will be some performance loss involved. However, from simulation tests run by the inventors, there will be less than 0.1 dB degradation caused by symmetry-based simplification for all the four constellations.

[0081] FIG. **9** illustrates a flowchart **900** of operations of the receiver shown in FIG. **3**. As shown in FIG. **9**, in step **910**, the demodulator **310** of FIG. **3** receives and demodulates a BIMC signal received over the channel **50** from the transmitter **100**, (noting that the BICM signal is encoded by the transmitter **100** via the LDPC encoder **110**). As discussed above, this Euclidean distance ED is calculated only for the quadrant that the received symbols are in and are used in step **930**, together with the SPLLR received from EI-based symbol probability generator **380** (which receives extrinsic information (EI) developed from the output of the LDPC decoder **360** of FIG. **3**) to generate the symbol a posteriori probability (APP) using the symbol APP generator **330**.

[0082] Still referring to FIG. **9**, in step **940**, the bit-LLR soft decision generator (SD) **340** generates soft decisions based on the symbol APP received from the symbol APP generator **330**. In step **950**, these bit-LLR soft decisions are converted into a priori probability LLR soft decisions by passing the output of the bit-LLR generator **340** through the first subtractor **345**, which subtracts the extrinsic information EI derived from the bit a posteriori probability LLR output of the LDPC decoder **360** from the output of the bit-LLR generator **340**, and deinterleaving the output of a first subtractor **345** to provide the bit a priori probability LLR soft decision which is input to the LDPC decoder **360** (see step **960**). As also shown in step **960**, the a priori probability LLR soft decision input to the LDPC decoder **360** is used by the LDPC decoder **360** to generate both information bit estimates of the received BICM signal from the transmitter **100** as well as to generate a posteriori probability (APP) LLR outputs from the decoder **360** to be used as feedback signals for performing the next iteration of processing. In step **970**, these APP LLR outputs of the LDPC decoder **360** are converted to extrinsic information (EI) by the operations of the second subtracter **365** and the interleaver **370**, as described above. In step **980**, the SPLLR is generated by the EI-based SPLLR generator **380** and fed to the symbol APP generator **330** (step **930**), as also described above, to perform the next iteration of processing the received BICM signal until the termination criteria is reached.

[0083] FIG. **10** is a block diagram showing an example a computer system **1000** upon which aspects of this disclosure may be implemented. The computer system **1000** may include a bus **1002** or other communication mechanism for communicating information, and a processor **1004** coupled with the bus **1002** for processing information. The computer system **1000** may also include a main memory **1006**, such as a random-access memory (RAM) or other dynamic storage device, coupled to the bus **1002** for storing information and instructions to be executed by the processor **1004**. The main memory **1006** may also be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor **1004**. The computer system **1000** may implement, for example, the BICM-ID systems described in FIGS. **1**-**9**.

[0084] The computer system **1000** may further include a read-only memory (ROM) **1008** or other static storage device coupled to the bus **1002** for storing static information and instructions for the processor **1004**. A storage device **1010**, such as a flash or other non-volatile memory may be coupled to the bus **1002** for storing information and instructions.

[0085] The computer system **1000** may be coupled via the bus **1002** to a display **1012**, such as a liquid crystal display (LCD), for displaying information. One or more user input devices, such as the example user input device **1014** may be coupled to the bus **1002**, and may be configured for receiving various user inputs, such as user command selections and communicating these to the processor **1004**, or to the main memory **1006**. The user input device **1014** may include physical structure, or virtual implementation, or both, providing user input modes or options, and a cursor control **1016** for controlling, for example, a cursor, visible to a user through display **1012** or through other techniques, and such modes or operations may include, for example virtual mouse, trackball, or cursor direction keys.

[0086] The computer system **1000** may include respective resources of the processor **1004** executing, in an overlapping or interleaved manner, respective program instructions. Instructions may be read into the main memory **1006** from another machine-readable medium, such as the storage device **1010**. In some examples, hard-wired circuitry may be used in place of or in combination with software instructions. The term “machine-readable medium” as used herein refers to any medium that participates in providing data that causes a machine to operate in a specific fashion. Such a medium may take forms, including but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media may include, for example, optical or magnetic disks, such as storage device **1010**. Transmission media may include optical paths, or electrical or acoustic signal propagation paths, and may include acoustic or light waves, such as those generated during radio-wave and infra-red data communications, that are capable of carrying instructions detectable by a physical mechanism for input to a machine.

[0087] The computer system **1000** may also include a communication interface **1018** coupled to the bus **1002**, for two-way data communication coupling to a network link **1020** connected to a local network **1022**. The network link **1020** may provide data communication through one or more networks to other data devices. For example, the network link **1020** may provide a connection through the local network **1022** to a host computer **1024** or to data equipment operated by an Internet Service Provider (ISP) **1026** to access through the Internet **1028** a server **1030**, for example, to obtain code for an application program.

[0088] While various embodiments have been described, the description is intended to be exemplary, rather than limiting, and it is understood that many more embodiments and implementations are possible that are within the scope of the embodiments.

[0089] In the following, further features, characteristics and advantages of the instant application will be described by means of items: [0090] Item 1: A receiver is provided for receiving and processing bit-interleaved coded modulation (BICM) signals from a BICM transmitter to generate information bit estimates of information in the BICM signals, the receiver including a decoder configured to generate the information bit estimates of the information in the received BICM signals, a symbol a posteriori probability (APP) generator configured to generate first symbol a posteriori probabilities (APPs) by processing the BICM signals based on Euclidean distances computed only for a quadrant to which received symbols derived from the BICM signals belong, and further based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator, wherein the extrinsic-information-based SPLLR generator is configured to generate the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), and the decoder is configured to generate the information bit estimates based on the first symbol APPs output from the symbol APP generator. [0091] Item 2: The receiver of item 1, wherein the receiver further comprises a bit-LLR soft decision generator configured to receive the first symbol APPs, and to generate bit-LLR soft decisions to provide to the decoder based on the first symbol APPs received from the symbol APP generator. [0092] Item 3: The receiver of items 1 or 2, wherein the decoder is configured to generate updated APPs, and further comprising a converter to convert the updated APPs to the extrinsic information to be provided to the SPLLR generator. [0093] Item 4: The receiver of any one of items 1 to 3, wherein the symbol APP generator, the extrinsic-information-based SPLLR generator, and the bit-LLR soft decision generator form a demapper for demapping the received BICM signals using a two-dimensional (2D) constellation index having M-complex symbols, and wherein the extrinsic-information-based SPLLR generator only generates the SPLLR signals for bits equal to zero in the 2D constellation index. [0094] Item 5: The receiver of any one of items 1 to 4, wherein the bit-LLR soft decision generator is configured to calculate the bit-LLR soft decisions using the SPLLRs. [0095] Item 6: The receiver of any one of items 1 to 5, wherein the receiver further comprises a demodulator configured to receive and demodulate the BICM signals from the transmitter, and wherein the demapper further comprises a Euclidean distance calculator configured to receive the demodulated BICM signals from the demodulator, and to convert the demodulated BICM signals into Euclidean distances to provide to the symbol APP generator. [0096] Item 7: The receiver of any one of items 1 to 6, wherein Euclidean distance calculator is configured to only calculate the Euclidean distances from one quadrant of the 2D constellation index. [0097] Item 8: The receiver of any one of items 1 to 7, wherein a priori probability LLRs are provided as an input to the decoder based on the bit-LLR soft decisions output from the bit-LLR soft decision generator. [0098] Item 9: The receiver of any one of items 1 to 8, wherein the converter comprises a subtracter configured to convert the updated APPs output from the decoder into the extrinsic information by subtracting the bit a priori probability LLR input to the decoder from the updated APPs output from the decoder. [0099] Item 10: The receiver of any one of items 1 to 9 The receiver of claim **1**, further comprising a deinterleaver located between the soft decisions output from the bit-LLR soft decision generator and the a priori probability input to the decoder, and an interleaver located between a subtractor and the extrinsic-information-based SPLLR generator. [0100] Item 11: The receiver of any one of items 1 to 10, wherein the decoder is a low-density parity check code decoder. [0101] Item 12: The receiver of any one of items 1 to 11, wherein for a first outer iteration for each received symbol, the quadrant is determined by the received symbol, and for the rest of the iteration the quadrant is determined by a maximum extrinsic based symbol probability. [0102] Item 13: A method for receiving and processing bit-interleaved coded modulation (BICM) signals from a BICM transmitter to generate information bit estimates of information in the BICM signals, the method comprising generating, via a decoder, the information bit estimates of the information in the received BICM signals, generating, via a symbol a posteriori probability (APP) generator, first symbol a posteriori probabilities (APPs) by processing the BICM signals based on Euclidean distances computed only for a quadrant to which received symbols derived from the BICM signals belong, and further based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator, wherein the extrinsic-information-based SPLLR generator is configured to generate the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), and the decoder is configured to generate the information bit estimates based on the first symbol APPs output from the symbol APP generator. [0103] Item 14: The method of item 13, further comprising receiving, via a bit-LLR soft decision generator, the first symbol APPs, and generating bit-LLR soft decisions to provide to the decoder based on the first symbol APPs received from the symbol APP generator. [0104] Item 15: The method of items 13 or 14, via the decoder, updated APPs, and converting, via a converter, the updated APPs to the extrinsic information to be provided to the SPLLR generator. [0105] Item 16: The method of any one of items 13 to 15, wherein the symbol APP generator, the extrinsic-information-based SPLLR generator, and the bit-LLR soft decision generator form a demapper for demapping the received BICM signals using a two-dimensional (2D) constellation index having M-complex symbols, and wherein the extrinsic-information-based SPLLR generator only generates the SPLLR signals for bits equal to zero in the 2D constellation index. [0106] Item 17: The method of any one of items 13 to 16, further comprising calculating, via the bit-LLR soft decision generator, the bit-LLR soft decisions using the SPLLRs. [0107] Item 18: The method of any one of items 13 to 17, further comprising receiving and demodulating, via a demodulator, the BICM signals from the transmitter, and receiving, via a Euclidean distance calculator in the demapper, the demodulated BICM signals from the demodulator to convert the demodulated BICM signals into Euclidean distances to provide to the symbol APP generator. [0108] Item 19: The method of any one of items 13 to 18, further comprising only calculating, via the Euclidean distance calculator, the Euclidean distances from one quadrant of the 2D constellation index. [0109] Item 20: The method of any one of items 13 to 19, wherein a priori probability LLRs are provided as an input to the decoder based on the bit-LLR soft decisions output from the bit-LLR soft decision generator.

[0110] Although many possible combinations of features are shown in the accompanying figures and discussed in this detailed description, many other combinations of the disclosed features are possible. Any feature of any embodiment may be used in combination with or substituted for any other feature or element in any other embodiment unless specifically restricted. Therefore, it will be understood that any of the features shown and/or discussed in the present disclosure may be implemented together in any suitable combination. Accordingly, the embodiments are not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

[0111] While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that the teachings may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all applications, modifications and variations that fall within the true scope of the present teachings.

[0112] Unless otherwise stated, all measurements, values, ratings, positions, magnitudes, sizes, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. They are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain.

[0113] The scope of protection is limited solely by the claims that now follow. That scope is intended and should be interpreted to be as broad as is

consistent with the ordinary meaning of the language that is used in the claims when interpreted in light of this specification and the prosecution history that follows and to encompass all structural and functional equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirement of Sections 101, 102, or 103 of the Patent Act, nor should they be interpreted in such a way. Any unintended embracement of such subject matter is hereby disclaimed.
[0114] Except as stated immediately above, nothing that has been stated or illustrated is intended or should be interpreted to cause a dedication of any component, step, feature, object, benefit, advantage, or equivalent to the public, regardless of whether it is or is not recited in the claims.
[0115] It will be understood that the terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein.
[0116] Relational terms such as first and second and the like may be used solely to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms "comprises," "comprising," or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element proceeded by "a" or "an" does not, without further constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
[0117] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claims require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

## Claims

**1**. A receiver in a receiver-based interference cancellation satellite system, including a bit-interleaved coded modulation (BICM) transmitter for transmitting signals to the receiver over a satellite link, for receiving and processing bit-interleaved coded modulation (BICM) signals from the BICM transmitter to generate information bit estimates of information in the BICM signals to improve power efficiency of the satellite link, the receiver comprising: a decoder configured to generate the information bit estimates of the information in the received BICM signals; a symbol a posteriori probability (APP) generator configured to generate first symbol a posteriori probabilities (APPs) by processing the BICM signals based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator, wherein: the extrinsic-information-based SPLLR generator is configured to generate the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), the decoder is configured to generate the information bit estimates based on the first symbol APPs output from the symbol APP generator, the receiver further comprises a bit-LLR soft decision generator configured to receive the first symbol APPs, and to generate simplified bit-LLR soft decisions to provide to the decoder based on the first symbol APPs received from the symbol APP generator, and the bit-LLR soft decision generator is configured to calculate the simplified bit-LLR soft decisions using the SPLLRs to improve the power efficiency of the satellite link with the simplified bit-LLR soft decisions.
**2**. The receiver of claim 1, wherein the decoder is configured to generate updated APPs, and further comprising a converter to convert the updated APPs to the extrinsic information to be provided to the SPLLR generator.
**3**. The receiver of claim 2, wherein the symbol APP generator, the extrinsic-information-based SPLLR generator, and the bit-LLR soft decision generator form a demapper for demapping the received BICM signals using a two-dimensional (2D) constellation index having M-complex symbols, and wherein the extrinsic-information-based SPLLR generator only generates the SPLLR signals for bits equal to zero in the 2D constellation index.
**4**. The receiver of claim 3, wherein the receiver further comprises a demodulator configured to receive and demodulate the BICM signals from the transmitter, and wherein the demapper further comprises a Euclidean distance calculator configured to receive the demodulated BICM signals from the demodulator, and to convert the demodulated BICM signals into Euclidean distances to provide to the symbol APP generator.
**5**. The receiver of claim 4, wherein Euclidean distance calculator is configured to only calculate the Euclidean distances from one quadrant of the 2D constellation index.
**6**. The receiver of claim 5, wherein a priori probability LLRs are provided as an input to the decoder based on the bit-LLR soft decisions output from the bit-LLR soft decision generator.
**7**. The receiver of claim 4, wherein the converter comprises a subtracter configured to convert the updated APPs output from the decoder into the extrinsic information by subtracting the bit a priori probability LLR input to the decoder from the updated APPs output from the decoder.
**8**. The receiver of claim 6, further comprising a deinterleaver located between the soft decisions output from the bit-LLR soft decision generator and a priori probability LLRs input to the decoder, and an interleaver located between a subtractor and the extrinsic-information-based SPLLR generator.
**9**. The receiver of claim 1, wherein the decoder is a low-density parity check code decoder.
**10**. The receiver of claim 1, wherein for a first outer iteration for each received symbol, a quadrant is determined by the received symbol, and for other iterations of for each received symbol, the quadrant is determined by a maximum extrinsic based symbol probability.
**11**. A method using a receiver-based interference cancellation satellite system, including a bit-interleaved coded modulation (BICM) transmitter for transmitting signals to a receiver over a satellite link, for receiving and processing bit-interleaved coded modulation (BICM) signals from the BICM transmitter to generate information bit estimates of information in the BICM signals to improve power efficiency of the satellite link, the method comprising: generating, via a decoder, the information bit estimates of the information in the received BICM signals; generating, via a symbol a posteriori probability (APP) generator, first symbol a posteriori probabilities (APPs) by processing the BICM signals based on symbol probability log-likelihood ratios (SPLLRs) provided to the symbol APP generator by an extrinsic-information-based symbol probability log-likelihood ratio (SPLLR) generator, wherein: the extrinsic-information-based SPLLR generator is configured to generate the SPLLRs directly from extrinsic information based on updated symbol APPs output from the decoder, without converting the extrinsic information into log-likelihoods (LLs), and the decoder is configured to generate the information bit estimates based on the first symbol APPs output from the symbol APP generator, the receiver further comprises a bit-LLR soft decision generator configured to receive the first symbol APPs, and to generate simplified bit-LLR soft decisions to provide to the decoder based on the first symbol APPs received from the symbol APP generator, and the bit-LLR soft decision generator is configured to calculate the simplified bit-LLR soft decisions using the SPLLRs to improve the power efficiency of the satellite link with the simplified bit-LLR soft decisions.
**12**. The method of claim 11, generating, via the decoder, updated APPs, and converting, via a converter, the updated APPs to the extrinsic information to be provided to the SPLLR generator.
**13**. The method of claim 12, wherein the symbol APP generator, the extrinsic-information-based SPLLR generator, and the bit-LLR soft decision generator form a demapper for demapping the received BICM signals using a two-dimensional (2D) constellation index having M-complex symbols, and wherein the extrinsic-information-based SPLLR generator only generates the SPLLR signals for bits equal to zero in the 2D constellation index.
**14**. The method of claim 13, further comprising receiving and demodulating, via a demodulator, the BICM signals from the transmitter, and receiving, via a Euclidean distance calculator in the demapper, the demodulated BICM signals from the demodulator to convert the demodulated

BICM signals into Euclidean distances to provide to the symbol APP generator.
**15**. The method of claim 14, further comprising only calculating, via the Euclidean distance calculator, the Euclidean distances from one quadrant of a 2D constellation index.
**16**. The method of claim 15, wherein a priori probability LLRs are provided as an input to the decoder based on the bit-LLR soft decisions output from the bit-LLR soft decision generator.